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Line spectrum tracking method in transformed signal space for underwater moving targets in low signal-to-noise ratio environment

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ABSTRACT:

Line spectrum tracking is an essential signal-processing method for underwater passive detection. However, its performance is often seriously degraded due to signal fluctuation, especially in low signal-to-noise ratio scenarios. In this paper, based on signal space transformation and hidden Markov model, a signal trajectory tracking method is proposed for underwater moving target detection and parameter estimation. With this method, tracking the varying line spectrum trajectory in three-dimensional frequency-azimuth-time signal space is constrained onto a two-dimensional plane in the transformed signal space. Not only is the computation complexity reduced, but the ability to track weak line spectrums and estimate target parameters is improved. The performance of this method is verified with numerical simulations and experimental data processing. © 2024 Acoustical Society of America. .

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I. INTRODUCTION

Low-frequency line spectrum (LS) components in radiation noise emitted by many underwater targets, known as tonal signals, contain vital information and exhibit high intensity and stability.¹ Therefore, low-frequency LS tracking is a crucial task for underwater passive target detection, especially for low signal-to-noise ratio (SNR) situations.

Traditional detect-before-track (DBT) methods perform signal detection on individual frame data before tracking and can be implemented within the low-frequency analysis and recording (LOFAR) spectra. In order to enhance the processing gain, array beamforming methods are widely used, and tracking is often achieved on the bearing time recording (BTR) spectrum as well. However, the array gain is limited by the array aperture size and element number. Various methods have been explored to improve frequency and azimuth resolution in LOFAR and BTR figures, such as deconvolution methods for power spectra² and conventional beamforming.³ Although increasing the number of snapshots can also improve signal-processing gains, the temporal variations in frequency and azimuth caused by target motion may affect the performance of time-frequency analysis and array signal processing.

Consequently, to further accommodate lower SNR, the concept of track-before-detect (TBD) is introduced, which

focuses on combining information from the multi-frame signal. TBD can be implemented in two main ways: treating it as a detection problem or considering it as a tracking problem.⁴ Algorithms focusing on detection aim to find a trajectory that maximizes energy accumulation over a long period. For instance, the TBD algorithms based on Hough transform⁵ accumulate energy along straight-line spectra in LS detection. However, its effectiveness is limited by non-linear changes in signal trajectory caused by target motion. In approaches treating TBD as a tracking problem, recursive Bayesian filtering technology is commonly employed. It continuously predicts and updates the probability distribution of the target state based on observations and prior knowledge, aiming to track a trajectory that maximizes the posterior probability. Numerous implementation methods, including Kalman filtering,⁶ extended Kalman filtering,⁷ unscented Kalman filtering,⁸ particle filtering,⁹ and others, have been developed. Since Barniv¹⁰ proposed the idea of discretizing state space, many dynamic programming algorithms, such as the forward-backwards algorithm¹¹ and Viterbi algorithm,¹² have been used for LS trajectory tracking. These methods, benefiting from the inherent discretization of frequency and azimuth bins following Fourier transformation in the time domain or spatial domain, are well-suited for frequency tracking¹³ and target azimuth trajectory extraction¹⁴ in passive sonar signal processing. The Bayesian filtering-based methods mentioned above demonstrate robust trajectory tracking capabilities under low SNR

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situations. However, achieving precise tracking for most algorithms requires SNR conditions where the LS trajectory can at least be faintly observed within the BTR or LOFAR figures.

Due to the potential of additional dimension of information to enhance the ability to suppress interference, some space–frequency joint LS detection and tracking methods have been proposed. For example, active sonar and radar have been able to utilize three-dimensional (3D) signal spaces, such as range–azimuth–time or range–Doppler–time, for trajectory tracking maturely. However, passive sonar faces challenges in tracking weak trajectories in frequency–azimuth–time (FRAZ-T) signal space due to limited prior information about targets, resulting in fewer existing methods. In 1991, Maranda and Fawcett¹⁵ introduced a technique for optimal LS trajectory search in multi-frame frequency–azimuth (FRAZ) spectra, requiring searching for five parameters to determine a curve in signal space for a target with uniform motion. This method relies on exhaustive search, demanding significant computational resources. Fawcett and Maranda¹⁶ and Maranda and Fawcett¹⁷ subsequently improved the computational speed using methods such as the Newton method with quadratic convergence and simulated annealing. Sun and Wang¹⁸ further refined the approach using particle swarm optimization (PSO) for iterative parameter search. However, the computational complexity of searching for a 3D space curve remains a limitation. In 2021, Luo and Shen¹⁹ improved LS tracking under low-SNR conditions by modeling the FRAZ-T as a hidden Markov model (HMM) and using the Viterbi algorithm to estimate frequency and azimuth states, utilizing the continuity of the spectrum between multiple frames. However, the high computational complexity of the Viterbi algorithm for a two-dimensional HMM, along with strong interference from neighboring frequencies and azimuths, impacts trajectory tracking performance. Additionally, Wang *et al.*²⁰ utilized convolutional neural networks to process multi-beam LOFAR spectra for passive underwater target tracking, showing the potential of deep learning for passive sonar target detection.

As is well known, uniform linear motion is a common strategy for oceanic targets like ships, as it ensures stability and energy efficiency by minimizing eddy currents and turbulence. Hence, an LS tracking method focusing on the low-frequency harmonic signal radiated by underwater targets undergoing uniform linear motion is proposed. By introducing the signal space transformation (SST) method²¹ based on the Doppler frequency formula, linear constraints were added for trajectory tracking. The PSO algorithm²² is used to search for three target parameters, and during the signal state tracking, the target parameters are estimated simultaneously. It is shown that compared with existing space–frequency joint methods, the proposed method achieves trajectory tracking under lower SNR conditions within a limited computing time.

The paper is organized as follows: in Sec. II, the model of an array receiving signals and the FRAZ-T 3D signal

space is established, the SST method is introduced, and the theoretical basis for the proposed method is given by discussing two existing trajectory tracking methods. Section III presents the main steps of the LS detection method based on trajectory tracking in the transformed domain, along with specific details. Section IV compares the performance of the proposed algorithm with existing space–frequency joint algorithms in simulations, focusing on trajectory tracking, parameter optimization, and system complexity. In Sec. V, the effectiveness of the proposed algorithm is validated by processing horizontal array data obtained in the SWellEx-96 experiment. Section VI provides a summary of the paper.

II. STATEMENT OF PROBLEM

In this section, the received signal model of a horizontal array and the FRAZ-T 3D signal space is established. The SST method is introduced to deal with the LS tracking problem in low SNR.

A. Array signal modeling and signal space construction

As shown in Fig. 1, a uniform linear array of M sensors is placed along the x axis with its center at the origin O to receive signals radiated by a target moving with constant velocity $\vec{v}$. When the target moves from point C to D , its distance relative to the original point changes from $\vec{r}_0$ to $\vec{r}(t)$. The angular frequency of the continuous tonal signal emitted during the target's movement is $\omega_0 = 2\pi f_0$. Assuming the distance between the target and the array is significantly greater than each other's aperture, then the target and the array are located in each other's far field. In such situations, the received signal arriving at an angle $\alpha_s(t)$ can be approximated as reaching the array in the form of a plane wave,²³ and the amplitude A_m of the received signal for each sensor is consistent.

The signal received by the m th sensor of the array at $(x, 0)$ can be expressed as

$$s(t, x) = A_m e^{i(\omega_0 t - \vec{k} \cdot \vec{r}(t) + k x \cos \alpha_s(t))} + n_r(t, x), \quad (1)$$

where $\vec{k}$ is the wavenumber vector, $k = |\vec{k}| = \omega_0/c$ is its modulus, with c being the sound speed in the water, and $n_r(t, x)$ is the sensor received background noise. The geometric relationship shown in Fig. 1 leads to

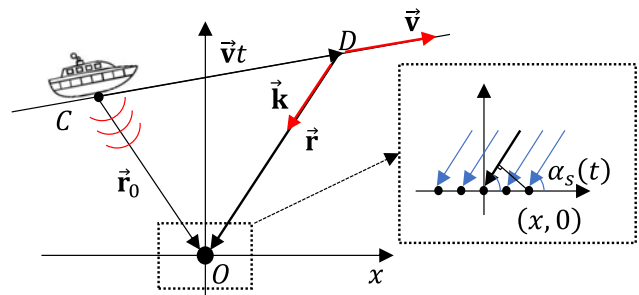


FIG. 1. (Color online) Diagram of linear array receiving target radiation.

$$s(t, x) = A_m e^{-i(\vec{k} \cdot \vec{r}_0 - (\omega_0 + \vec{k} \cdot \vec{v})t - kx \cos \alpha_s(t))} + n_r(t, x). \quad (2)$$

After conducting time–frequency analysis and frequency-domain beamforming on the array signal shown in Eq. (2), we obtain Eq. (3), i.e.,

$$P(f, \alpha, t) = \int_{-L/2}^{+L/2} \int_{-\infty}^{+\infty} s(t', x) g^*(t' - t) e^{-i2\pi f(t' + x \cos \alpha/c)} dt' dx + N_r(f, \alpha, t), \quad (3)$$

where $g(t)$ represents the sliding window used for the short-time Fourier transform (STFT), and L is the length of the array. Since the spectral density is constant in the whole frequency domain for assumed Gaussian white noise, and it is uniformly distributed in space, too, the obtained spectrum $N_r(f, \alpha, t)$ for the noise at a specific time is uniformly distributed on the FRAZ spectrum. By combining Eqs. (2) and (3), letting the amplitude be equal to 1, we obtain the signal spectrum

$$P(f, \alpha, t) = e^{i\vec{k} \cdot \vec{r}_0} \text{sinc}\left(\frac{f_0}{c}(\cos \alpha - \cos \alpha_s(t))\right) \times \text{sinc}\{f - f_r(t)\} + N_r(f, \alpha, t) \quad (4a)$$

and

$$f_r(t) = f_0 + \frac{\vec{k} \cdot \vec{v}}{2\pi} = f_0 + f_0 \frac{v}{c} \cos \theta(t), \quad (4b)$$

where f_r is the frequency of the received signal, and θ is the joint angle between $\vec{k}$ and $\vec{v}$, indicating that the frequency of the received signal conforms to the Doppler effect and is related to the radial velocity of the sound source. Equation (4) shows that the spectrum P reaches its maximum when the azimuth angle α and frequency f correspond, respectively, to the actual azimuth angle of the target and receiving signal frequency at a given time t . The power spectrum values Psd are obtained by taking the square of the spectral values P , and the signal space FRAZ-T is composed of power spectral values at different azimuths, frequencies, and times. As shown in Fig. 4(a), during the process of movement, especially when passing through the closest point of approach (CPA) relative to the array, the target may form a trajectory that changes nonlinearly with time in the FRAZ-T 3D signal space, passing through multiple beams and frequency bins.

B. Problem of LS trajectory extraction in 3D signal space

The traditional DBT methods extract the azimuth and frequency information of the signal on the FRAZ spectrum at each moment, which has certain requirements for SNR. The TBD methods can obtain the processing gain brought by energy accumulation after searching or tracking the optimal trajectory. However, upon analyzing the

distribution of trajectory points in the signal space (as detailed in Sec. II A), it is evident that the extraction of nonlinear LS trajectory faces considerable challenges in the case of low SNR and lack of prior information of the target motion.

One of the current solutions is the space–time integration (STI)^{15–18} method, in which a parameter optimization algorithm is used to search for the five parameters determining the trajectory of a uniformly moving target in the 3D signal space, i.e., signal transmission frequency, target velocity, target moving course, initial distance, and initial azimuth. The cost function is established by integrating power spectral values along a trajectory defined by these parameters, with the optimal trajectory identified by maximizing the cost function using a parameter optimal algorithm, such as PSO, which has a robust global search capability and is easy to implement and adapts well to nonlinear and nonconvex problems.²² The above method uses prior knowledge of the uniform-motion model, significantly improving the processing gain via long-time incoherent integration. Theoretically, it has a low SNR requirement. Still, due to the need to search in a five-dimensional parameter space, the computational time required to obtain ideal results under low SNR is unbearable in practical applications.

Another kind of solution is based on HMM, such as the method shown in Ref. 19, in which the hidden state at time frame t is represented as $q_t = [f_t, \alpha_t]$, and the FRAZ spectrum Psd serves as the observed value z_t . The determination of the state transition probability is based on the prior knowledge that the frequency and azimuth angle, respectively, change continuously with time. The tracking of the state q_t in Ref. 19 is implemented using the Viterbi algorithm, which is a recursive optimal solution to the problem of maximum *a posteriori* probability estimating the state sequence of a finite-state discrete-time Markov process observed in memoryless noise.²⁴ In order to solve the problem of a large amount of calculation tracking in three-dimensional signal space, the author also proposes region segmentation and merging similar LS, which introduces more complex problems of determining which regions should be discarded and connecting line spectra in multiple regions. This method lacks the use of prior knowledge relevant to the target motion model.

To enhance the tracking performance in low-SNR situations and reduce the computational complexity, we introduce the correlation between state f_t and α_t when establishing the HMM. This is achieved using the SST method proposed in our previous work.²¹ In Sec. II C, a brief overview of this method is provided.

C. Signal space transformation

Figure 2 shows the geometric relationship between the azimuth angle α ($\alpha \in [0, 180^\circ]$) defined in the array coordinate system xOy and the relative bearing angle θ ($\theta \in [0, 180^\circ]$) defined in the target coordinate system $x'By'$. These angles

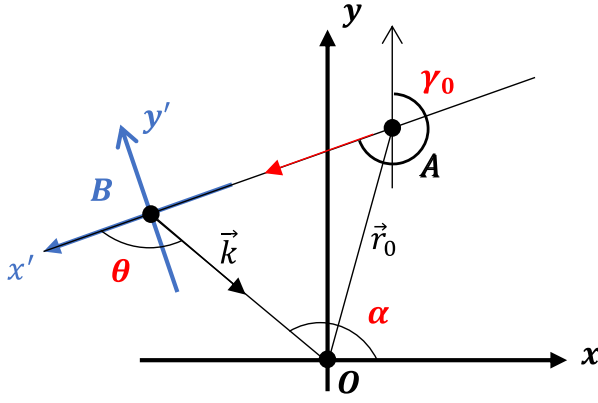


FIG. 2. (Color online) Diagram of target motion.

are connected via the target course angle γ_0 ($\gamma_0 \in [0, 360^\circ]$). Based on the above information and the Doppler effect revealed in Eq. (4b), after the following coordinate transformation, the 3D space FRAZ-T can be transformed to the signal space consisting of the frequency, the cosine value of the relative bearing, and the time:

$$\begin{cases} f' = f, \\ t' = t, \\ \cos \theta = -\sin(\alpha + \gamma_0). \end{cases} \quad (5)$$

In Ref. 21, we discussed two cases for azimuth axis transformation, which depend on the location of the CPA of the target trajectory relative to the original point O . As shown in Fig. 3(a), the target is represented by an asterisk, R_{CPA} is the distance between the CPA and the array, and the distance between the target and the CPA is represented by d . Our current discussion only considers the scenario in which the target moves in the first and second quadrants of the coordinate system xOy . During the coordinate transformation, the definition domain of azimuth angle α is based on the chosen course angle γ_0 . If the CPA lies to the left side of the y axis, then α is in the range $(\beta, 180^\circ)$; conversely, if the CPA is to the right side of the y axis, then α is in the range $(0^\circ, \beta)$; where $\beta = 90^\circ + n * 180^\circ - \gamma_0$, n is an integer, and β ranges from 0° to 180° .

Taking a course angle of 90° as an example, as shown in Fig. 4(a), the trajectory of the moving-target signal

received by the array in the 3D space FRAZ-T is represented as a 3D curve, indicated by the dotted line. Here, both the frequency and azimuth change nonlinearly with time. The signal trajectory in the transformed 3D space is illustrated in Fig. 4(b). When the target moves at a constant speed along a straight line, the relationship between $f(t)$ and $\cos \theta(t)$ is linear. The signal trajectory points are projected onto the frequency– $\cos \theta$ plane as a straight line. Thus, the trajectory in the transformed 3D signal space can be represented as a curve on slice I: $f - (f_0 + f_0(v/c)\cos \theta) = 0$. As shown in Fig. 4(b), slice I consists of the time axis t and the axis u and is determined by the parameters f_0 and v . The axis u in the slice represents the measurement under the joint action of the Doppler frequency shift and the change of the cosine value of the relative bearing, including information of the relationship between the azimuth and the frequency domain.²¹

Based on the previous discussion, we have identified the essential parameters for determining a slice in the space: $Q_s = [\gamma_0, f_0, v]$. Once these three parameters are selected, the two state variables f_t and α_t can be transformed into a single state variable u_t . Consequently, trajectory tracking in the 3D space is transformed into trajectory tracking on the 2D slice. The specific implementation scheme is described in detail in Sec. III.

III. LINE SPECTRUM TRACKING METHOD IN THE TRANSFORMED SIGNAL SPACE

The LS tracking method proposed herein combines SST, HMM, and the parameter optimization method (PSO), so the algorithm is abbreviated as PSO-SST-HMM. In this section, the procedure for implementing this algorithm is introduced.

A. Main steps

Figure 5 shows a flow chart of the algorithm implementation. First, time-frequency analysis and frequency-domain beamforming are applied to the signal received by the array, resulting in a multi-frame FRAZ spectrum, i.e., FRAZ-T 3D signal space. Second, within the set parameter interval, an initial set of moving target parameters γ_0 , f_0 , and v is randomly provided for the PSO iteration process. SST is performed based on parameter γ_0 , and then slice I is obtained in

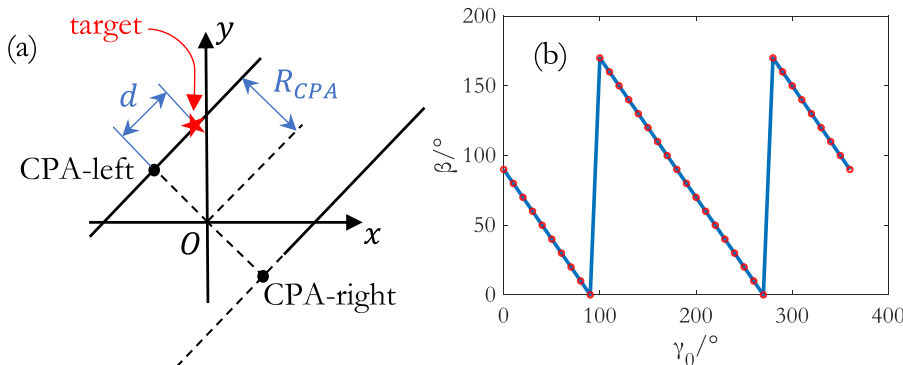


FIG. 3. (Color online) (a) Relationship between the closest point of approach (CPA) and y axis, (b) values of β for different target course angles.

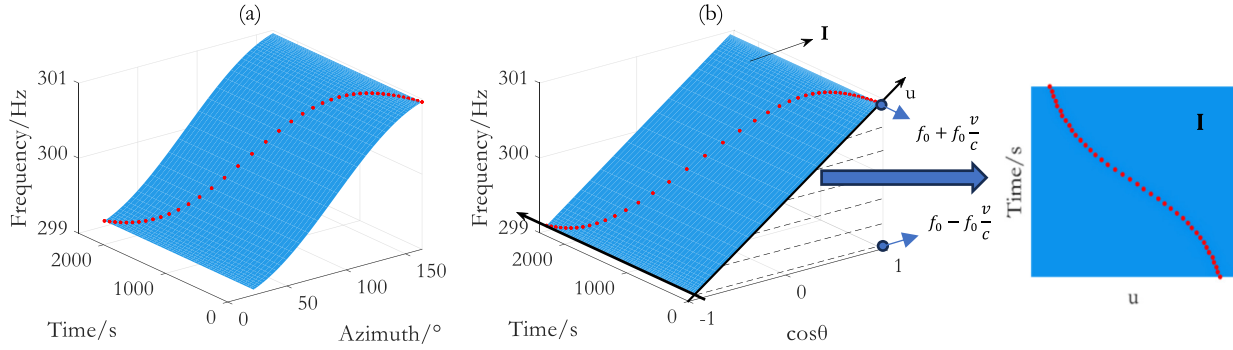


FIG. 4. (Color online) Distribution of trajectory points in (a) FRAZ-T 3D signal space, (b) frequency-cosθ-time 3D signal space.

the transformed space based on parameters f_0 and v . Third, the Viterbi algorithm is employed to track the signal trajectory on the obtained slice. Then, the power spectrum values of the effective trajectory points are accumulated to serve as the cost function value for the parameter set. If the termination criteria of the PSO algorithm are not met, the parameters are adjusted, and iteration continues until the best estimate of the target's parameters is obtained. Finally, the slice can be obtained with the optimal parameter values, and then target tracking can be achieved.

B. Construction of cost function

1. Trajectory tracking

The HMM is established by determining the hidden state, observed value, and three elements, namely $\{X, Z, \lambda\}$, where $\lambda = [\pi, A, B]$. The slice obtained in transformed space, as shown in Fig. 4(b), has a one-to-one mapping relationship between its u -axis and the frequency axis. So, when tracking LS on the slice, in order to make the physical meaning more transparent, the set of all finite states in HMM is set as $X = [f_l : df : f_h]$, which is the set of frequencies f rather than u , where f_l and f_h are the values closest to $(f_0 - f_0 v/c)$ and $(f_0 + f_0 v/c)$ in all frequency bins, respectively. The Fourier transform's frequency resolution is df , and the slice's bandwidth is approximately $2f_0 v/c$, usually a small value. If the target parameters are selected correctly, a single frequency LS is included within this narrow bandwidth, allowing the use of a single-LS tracking method. The finite measurement set is denoted as $Z = [z_1, \dots, z_t, \dots, z_T]$, where $z_t = P_t(X)$ is the power spectrum value corresponding to each frequency bin, and T is the number of time frames.

Because of the absence of prior values, the initial probability vector π is assumed to follow a uniform distribution. For a frequency range consisting of N intervals, we express π as

$$\pi_i = \frac{1}{N}, \quad i = 1, \dots, N. \quad (6)$$

The state transition probability is assumed to follow a Gaussian distribution with a mean value of zero and a variance of σ_f^2 , i.e.,

$$h_{ij} = P_r(q_k = i | q_{k-1} = j) = \frac{1}{\sigma_f \sqrt{2\pi}} \int_{f_j}^{f_{j+1}} \exp\left(-\frac{(f - \bar{f}_i)^2}{2\sigma_f^2}\right) df, \quad i, j > 0, |i - j| \leq f_{shift}, \quad (7)$$

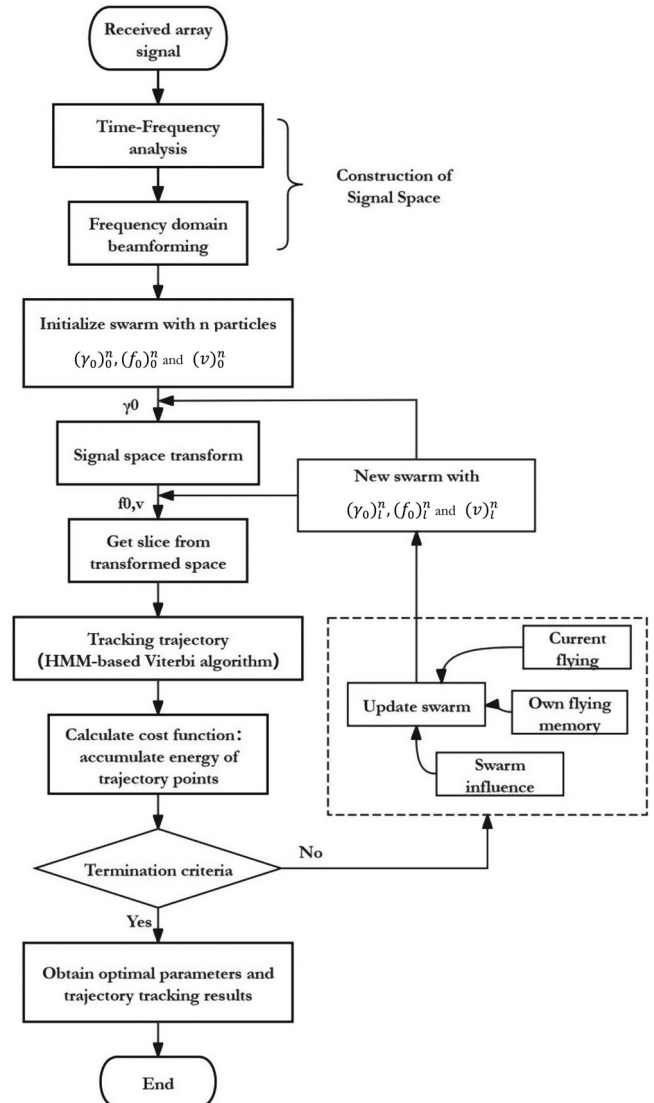


FIG. 5. Flow chart of the line spectrum (LS) tracking method.

where $\bar{f}_i$ is the center frequency of the i th frequency interval $[f_i, f_{i+1}]$, and f_{shift} is the maximum offset range of state i . The element a_{ij} in the state transition matrix $\mathbf{A}$ can be obtained by normalizing the matrix h_{ji} , i.e.,

$$a_{ji} = \frac{h_{ji}}{\sum_{n=1}^N h_{jn}}, \quad i, j = 1, \dots, N. \quad (8)$$

$\mathbf{B}$ is the measurement probability matrix, which can be obtained using the nonparametric integration method of the power spectrum when the SNR is unknown, i.e.,

$$b_i(z_t) = \frac{P_t(i)}{\sum_{j=1}^N P_t(j)}. \quad (9)$$

By using the Viterbi algorithm on the established model, as shown in the recursive Eq. (10), the tracked trajectory $\{\hat{q}_1, \hat{q}_2, \dots, \hat{q}_T\} = \{\hat{f}_1, \hat{f}_2, \dots, \hat{f}_T\}$ can be obtained.

$$\delta_1(i) = \frac{1}{N} b_i(z_1), \quad i = 1, \dots, N, \quad (10a)$$

$$\delta_t(i) = \max_j [\delta_{t-1}(j) a_{ji}] b_i(z_t), \quad (10b)$$

and

$$\hat{q}_t = \arg\max_j \{\delta_t(j)\}. \quad (10c)$$

2. Energy accumulation at effective trajectory points

Not all points obtained from the trajectory tracking mentioned above are correct. This occurs due to two main reasons. First, in some cases, there is no signal trajectory in the slices corresponding to certain parameter groups, leading the algorithm to track many scattered points. Second, in low SNR environments, the slice containing the signal trajectory can be affected by noise, causing the tracked trajectory points to deviate from the actual trajectory at some times. Therefore, it is necessary to determine whether the tracked points $\{\hat{q}_1, \hat{q}_2, \dots, \hat{q}_T\}$ are the effective trajectory points.

The hidden state ϕ is introduced to signify the presence or absence of the target, denoted as $H_1: \phi_t = 1$ and $H_0: \phi_t = 0$. The existence and nonexistence of the target in the initial probability vector are set to equal probability, i.e.,

$$\pi_i = \frac{1}{2}, \quad i = 1, 2. \quad (11)$$

The state transition matrix is defined as $\mathbf{A} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$. It is commonly assumed that the signal trajectory shows continuity in frequency, implying that the frequency value at a later moment is highly likely to be close to the value at the previous moment. In contrast, false trajectory points generated by noise lack this characteristic. The probability

density function of the frequency value at time t can be modeled as follows:²¹

$$\begin{cases} P_r(f_t | \phi_t = 0) = \frac{1}{cst}, \\ P_r(f_t | \phi_t = 1) = \frac{1}{\sqrt{2\pi\sigma_f^2}} \exp\left(-\frac{(f_t - \bar{f}_{t-1})^2}{\sigma_f^2}\right), \end{cases} \quad (12)$$

where cst is a constant greater than 1, and $\bar{f}_{t-1}$ is the average frequency of the previous moments. Thus, the likelihood function matrix $\mathbf{B}$ can be written as

$$\mathbf{B} = \begin{bmatrix} \rho & \frac{P_r(f_1 | \phi_1 = 1)}{P_r(f_1 | \phi_1 = 0)} \\ \vdots & \vdots \\ \rho & \frac{P_r(f_T | \phi_T = 1)}{P_r(f_T | \phi_T = 0)} \end{bmatrix}, \quad (13)$$

where ρ is the likelihood ratio detection threshold, which is calculated as

$$\rho = cst \times \frac{1}{\sqrt{2\pi\sigma_f^2}} \exp\left(-\frac{\Delta f_{max}^2}{\sigma_f^2}\right), \quad (14)$$

where Δf_{max} is the maximum frequency change Δf within the step length Δt of the sliding time window. The expression for Δf can be derived from the geometric relationship in Fig. 3(a), i.e.,

$$\Delta f = \frac{f_0 v}{c} \left(\frac{d}{\sqrt{d^2 + R_{CPA}^2}} - \frac{d - v * \Delta t}{\sqrt{(d - v * \Delta t)^2 + R_{CPA}^2}} \right). \quad (15)$$

As shown in Eq. (15), $d = 0$ corresponds to the scenario where the target is exactly at the CPA, resulting in the maximum value of Δf , i.e.,

$$\Delta f_{max} = \frac{f_0 v}{c} \left(\frac{v * \Delta t}{\sqrt{(v * \Delta t)^2 + R_{CPA}^2}} \right). \quad (16)$$

Based on the new HMM established above, the Viterbi algorithm is applied again to the tracked trajectory to determine the existence (birth) or nonexistence (death) of trajectory points. Next, as in Eq. (17), the power spectrum energy at the trajectory points identified as signals is integrated and used as the cost function for PSO, i.e.,

$$J_s(Q_s) = \sum_{t \in E} P_{sd}(\hat{f}_t), \quad (17)$$

where E is a set of time serial numbers of the trajectory points identified as “existence,” as mentioned above, and P_{sd} is the power spectrum value in the slice. Note that

alternative single-line spectral trajectory tracking methods can also be applied, with the described tracking method serving as a reference.

IV. SIMULATION ANALYSIS

In this section, the performance of the proposed algorithm is analyzed and compared with two existing algorithms through simulation. As shown in Fig. 6, the simulation scenario involves a linear array comprising 27 elements, spaced at intervals of 3.5 m, deployed to capture a tonal signal of 300 Hz emitted by a moving target. The target traverses with a velocity of 5 m/s (random disturbances with a standard deviation of 0.25 generated by normal distribution are added to the horizontal component v_x and the vertical component v_y of the velocity v , respectively), originating from coordinates $(-3500, 2000\text{ m})$ at a course angle of 70° for 20 min. Gaussian white noise is added to the received signal, which is sampled at 1000 Hz. In STFT and frequency-domain beamforming, the window length and hop length are set to 60 and 30 s, and the angular resolution is configured at 0.5° . This section mainly studies the impact of different signal-to-noise ratios (SNRs) on the algorithm; simulation experiments were conducted without considering ocean terrain, sound speed profiles, and the distance between the target and the array. This is because for passive sonar receiving single-frequency signals emitted by ocean targets, the factors mentioned above generally only result in amplitude fluctuations of the signal during processing time, with little impact on the measurement of the target's frequency and orientation. The SNR in this study is defined as

$$SNR = 10 \log_{10} \left(\frac{\text{average power of signal within 1 Hz}}{\text{average noise power within 1 Hz}} \right). \quad (18)$$

A. Performance of trajectory tracking

In order to evaluate and compare the tracking accuracy of the three algorithms under different SNR conditions, the trajectory point detection probability and the root mean square error (RMSE) of the two states are defined.

The trajectory point detection probability is defined as

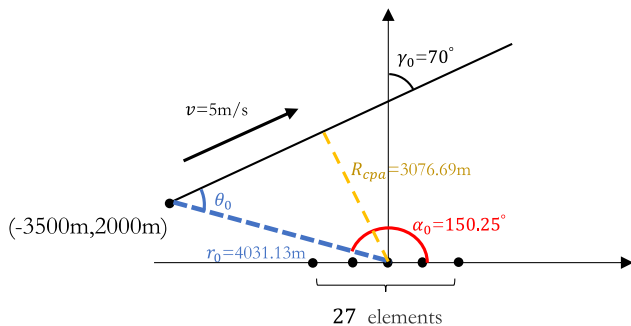


FIG. 6. (Color online) Schematic diagram of simulated target motion.

$$P_{D_{\text{trajectory}}} = \frac{Num_{\text{detected}}}{Num_{\text{total}}} \times 100\%, \quad (19)$$

where Num_{detected} is the number of correctly detected track points and Num_{total} is the actual total number of track points. A trajectory point is considered correctly detected if the difference between the estimated and actual trajectory points is simultaneously less than 0.1 Hz in frequency and less than 10° in azimuth.

The RMSE is defined as

$$RMSE_{\alpha} = \sqrt{\frac{1}{N_e} \sum_{t \in E} (\alpha_t - \hat{\alpha}_t)^2} \quad (20a)$$

and

$$RMSE_f = \sqrt{\frac{1}{N_e} \sum_{t \in E} (f_t - \hat{f}_t)^2}, \quad (20b)$$

where N_e is the total number of tracked trajectory points, f_t and α_t is the actual frequency and arrival angle of the received signal at time t , $\hat{f}_t$ and $\hat{\alpha}_t$ is the estimated frequency and arrival angle.

We conducted 100 Monte Carlo simulations on the simulated signals with SNR of -25 , -20 , -15 , and -5 dB using three algorithms, respectively. In the STI algorithm,^{15–18} we use the PSO algorithm to search for the five trajectory parameters, referred to herein as PSO-STI. As shown in Table I, the PSO algorithm was used in both PSO-STI and PSO-SST-HMM methods, and all processing parameters were set to be consistent except for different numbers of parameters that need to be searched. Due to the number of particles playing an important role in the search results, we conducted experiments with three different particle numbers at each SNR, namely 50, 500, and 5000. In addition, the space-frequency joint detection and tracking (SFJDT) algorithm, introduced in Ref. 19, establishes a 2D HMM in the 3D space for tracking the target trajectory. In simulation calculations, we define its state space as $f = [299 : 1/\Delta T : 301]$ and $\alpha = [0 : 0.5^\circ : 180^\circ]$, where ΔT represents the window length used during segmentation. However, for the HMM in the PSO-SST-HMM algorithm, the state is one-dimensional, as described in Sec. III B, $f = [f_l : 1/\Delta T : f_h]$, where the f_l and f_h depend on the selection of the parameter set. In this paper, the Viterbi algorithm was employed with a lookback length of 38 to cover the entire trellis for the simulated data with a total of 39 segments.

Figure 7 shows the LS tracking results of 100 simulated signals using the three algorithms under different SNRs, while Fig. 8 shows the impact of SNR on the average trajectory point detection rate and RMSE. It can be seen that PSO-SST-HMM outperforms PSO-STI and SFJDT in tracking tonal signals. It provides more precise target information in the frequency and azimuth dimensions, even with lower SNR and limited computing time. Specifically, it achieves accurate trajectory tracking with only 50 particles at SNRs above -20 dB and performs well with 500 particles at low SNRs of around -25 dB. In contrast, the PSO-STI method requires more

TABLE I. Selection of processing parameters in three algorithms.

		PSO-SST-HMM	PSO-STI	SFJDT
PSO	The range of parameters to be searched:	$f_0 \in [299\ 301]$ Hz $\gamma_0 \in [0^\circ, 180^\circ]$ $v \in [1, 10]$ m/s	$f_0 \in [299\ 301]$ Hz $\gamma_0 \in [0^\circ, 180^\circ]$ $v \in [1, 10]$ m/s $r_0 \in [600\ 08\ 000]$ m $\alpha_0 \in [0^\circ, 180^\circ]$	
	Other parameters:	Boundary handling: clamping boundaries Function tolerance: 1e-6 Inertia range: [0.1, 1.1] Maximum number of iterations: 20 000 Minimum adaptive neighborhood size: 0.25. Weighting of each particle's best position when adjusting the parameters: 1.49. Weighting of the neighborhood's best position when adjusting the parameters: 1.49. particle numbers: 50, 500, 5000		
HMM	Sets of finite states		$f = [f_l : 1/60 : f_h]$ Hz $299 \leq f_l \leq f_h \leq 301$	$f = [299 : 1/60 : 301]$ Hz $\alpha = [0 : 0.5^\circ : 180^\circ]$
	Other parameters:		Backtracking length: 38	

particles, and even 5000 particles are insufficient at an SNR of -25 dB. Based on this, this paper uses the PSO-SST-HMM algorithm with 500 particles and the PSO-STI algorithm with 5000 particles for performance comparison. However, when we set the two parameters r_0 and α_0 in the PSO-STI method as known variables and then use the PSO algorithm with 500

particles to search for the remaining three parameters $[f_0, v, \gamma_0]$, it achieves high trajectory detection rates under different SNRs, similar to the proposed method. Therefore, the limitations of the PSO-STI method are mainly due to its high parameter dimensionality and large search space, making finding the global optimal value challenging, especially in low

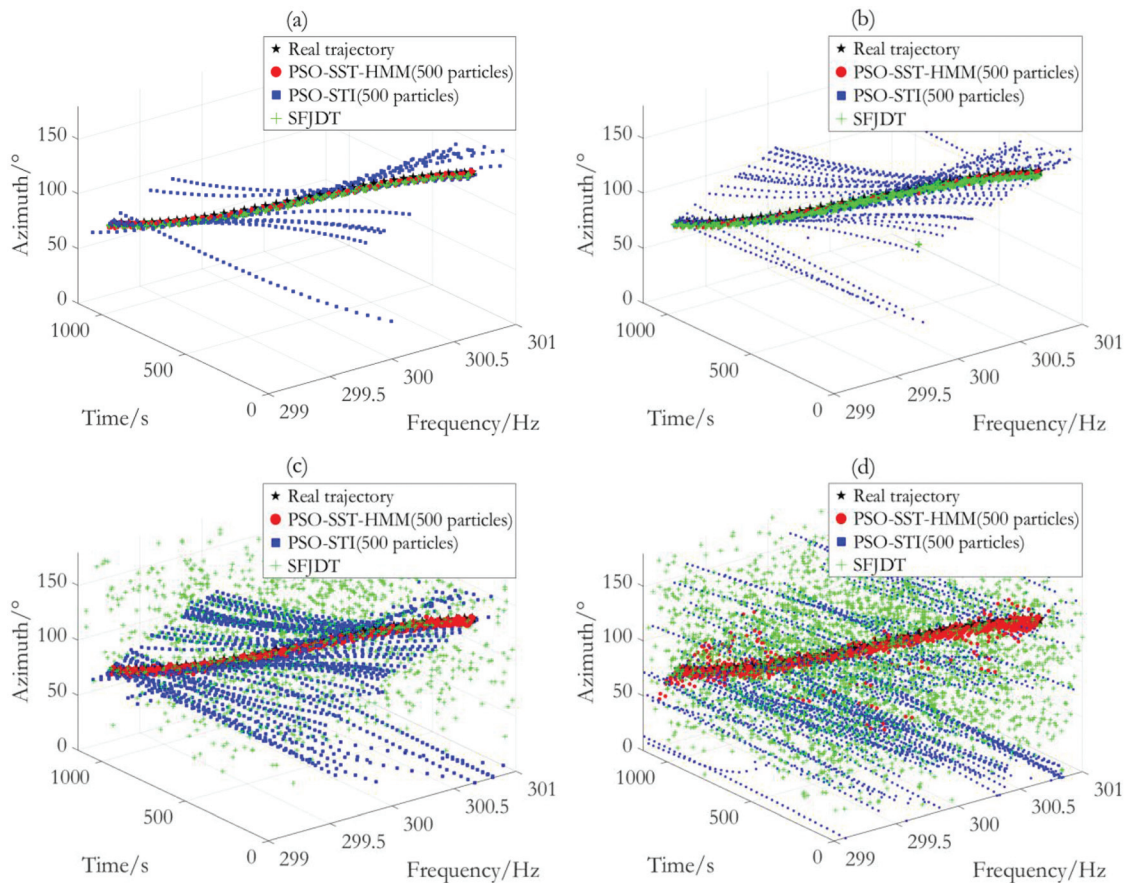


FIG. 7. Comparison of real trajectory and 3D space trajectories estimated by the three algorithms for SNR of (a) -5 n dB, (b) -15 dB, (c) -20 dB, (d) -25 dB.

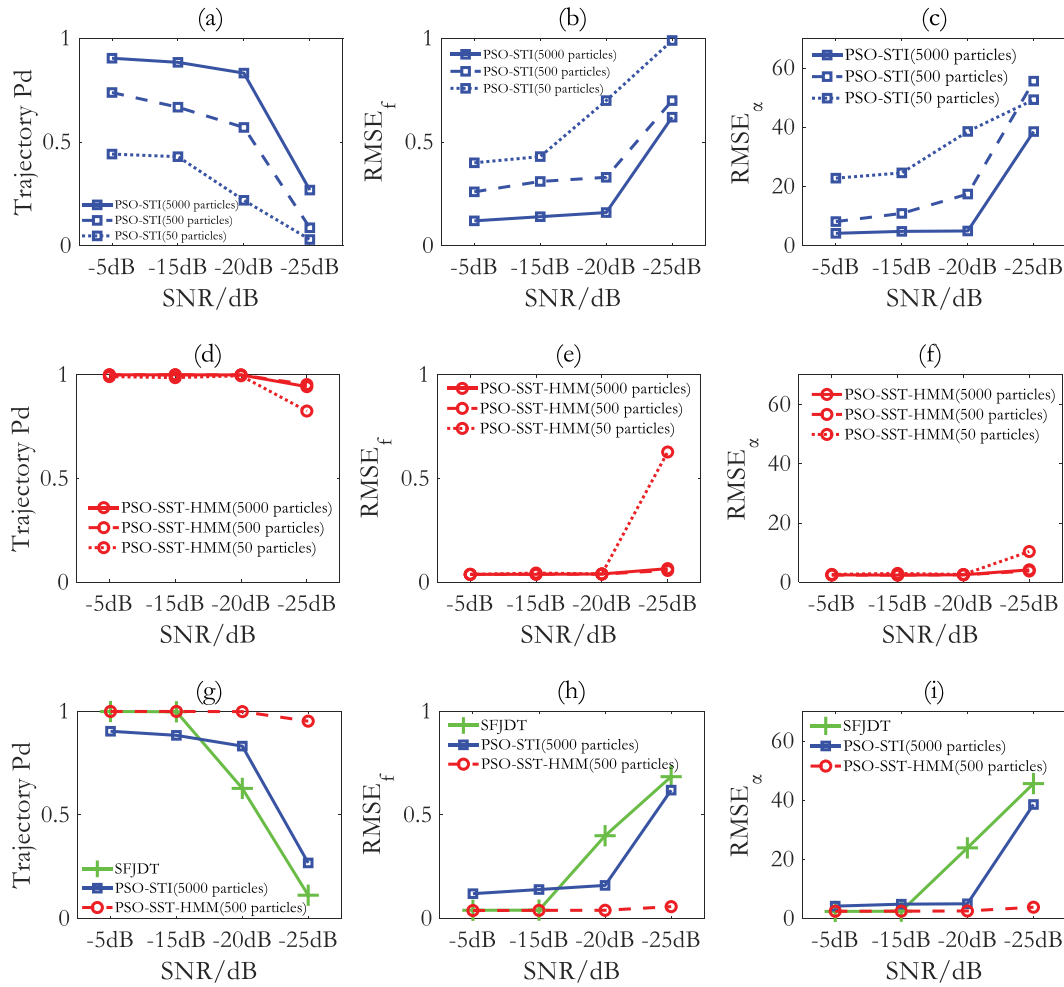


FIG. 8. (Color online) Influence of SNR on trajectory point detection probability and root mean square error of tracking: (a)–(c) performance of PSO-STI algorithm at different numbers of particles, (d)–(f) performance of PSO-SST-HMM algorithm at different numbers of particles, (g)–(i) performance comparison of three algorithms.

SNR conditions. PSO-SST-HMM achieved better performance by reducing the search dimension of parameters. This also indicates that the gain of PSO-SST-HMM here, as analyzed in Ref. 21, still mainly comes from beamforming, 60 s FFT, and the accumulation of multiple trajectory points. The PSO algorithm and Viterbi algorithm in this paper are both means used to achieve this gain efficiently.

The tracking performance of the SFJDT algorithm becomes poor when the SNR is below -20 dB. Compared with SFJDT, PSO-SST-HMM uses not only the stability of state change but also the information of target motion parameters. A complete signal trajectory exists on the slice corresponding to the correct parameter group, enabling tracking by the Viterbi algorithm. As shown in Fig. 9(a), in SFJDT, the state's position in the FRAZ spectrum at each time is determined during each iteration of the Viterbi algorithm. At time t , the number of states for which posterior probability needs to be calculated is $F_{bin} * A_{bin}$ (i.e., the number of frequency bins multiplied by the number of azimuth bins), representing the number of candidate states. In PSO-SST-HMM, using the Viterbi algorithm on the slice

reduces the selectable states to a few discrete points on the line segment $A_i B_i$ as shown in Fig. 9(b). The performance is significantly better for a dynamic programming algorithm with a lower number of possible transitions.²⁸ This is because the energy difference between the signal and noise points is relatively small at low SNR, and the energy and the state spacing determine the posterior probability. With more states, there is a higher chance of a noise point with energy exceeding the signal point. As a result, the likelihood of making incorrect judgments during the iterative process rises. Therefore, PSO-SST-HMM exhibits a higher trajectory detection rate under low SNR conditions than SFJDT.

B. Parameter optimization performance

To investigate the performance of the PSO-SST-HMM algorithm in parameter optimization, we presented the process of parameter iteration convergence. Moreover, the accuracy of parameter estimation was demonstrated by calculating the relative error between the estimated parameter

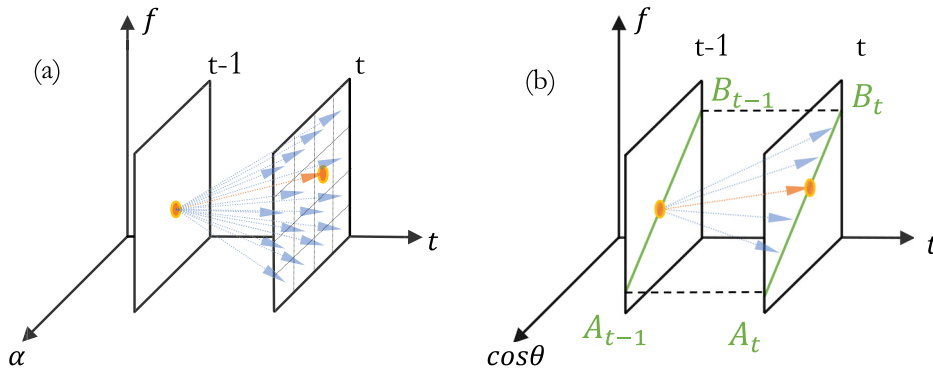


FIG. 9. (Color online) Schematic of state transition: (a) 2D HMM, (b) 1D HMM.

and the actual parameter when using particle number as a variable under different SNRs.

Figure 10 shows the convergence process of each target parameter using the PSO algorithm in the PSO-SST-HMM method when the SNR is -25 dB. The red dotted line represents the actual parameter values. After approximately 60 iterations, the target course γ_0 , emission frequency f_0 , and target speed v converged to 73.18° , 299.95 Hz, and 5.01 m/s, respectively, and the convergence results are satisfactory.

Figure 11 shows the relative error between the actual value and the mean of the 100 parameter estimation results obtained from the PSO-STI and PSO-SST-HMM algorithms at different SNRs. SFJDT algorithm does not estimate target parameters and thus cannot be compared here. Only results for three parameters shared with PSO-SST-HMM among PSO-STI's five parameters are displayed for comparison. The PSO-STI and PSO-SST-HMM algorithms must choose the appropriate

number of particles based on the SNR to obtain good parameter estimation results. It is observed that PSO-STI requires more particles than PSO-SST-HMM at all four SNRs. When the SNR is below -20 dB, even with 5000 particles, PSO-STI has a significant error in estimated parameters.

For PSO-SST-HMM, the accuracy of parameter estimation directly influences the trajectory tracking performance, as only the slice determined by three correct parameters contains the entire signal trajectory. When the SNR is higher than -25 dB, 500 particles are approximately sufficient for parameter estimation, and the relative error between the three estimated parameters and the actual values is less than 10%, corresponding to the high trajectory detection rate in Sec. IV A. When the SNR is below -25 dB, the performances of trajectory tracking and parameter estimation of this method synchronously decrease. This demonstrates the importance of constructing slices for trajectory tracking in the PSO-SST-HMM algorithm.

C. System complexity

In order to investigate the availability of the algorithm in practical applications, this section compares the computational complexity of the three algorithms and shows the operation time of simulation processing.

Figure 12 shows the average running time of 100 Monte Carlo calculations of three algorithms (PSO-STI, SFJDT, and PSO-SST-HMM) under various SNR conditions. Taking advantage of the fast-computing speed of the Viterbi algorithm implemented on HMM of 1D state space, the time for calculating the cost function in the PSO-SST-HMM algorithm is similar to that in the PSO-STI algorithm. Therefore, the running time is mainly determined by the number of manually set particles, equating to the number of cost functions calculated at each iteration. Based on the results presented in Sec. IV A, when dealing with low SNR conditions and ensuring good tracking performance, the PSO-SST-HMM algorithm should have a significantly shorter operation time compared to PSO-STI. In fact, it takes only around 5 min to process 40 min of data, making it a practical choice for real-world applications.

When using the Viterbi algorithm for trajectory tracking, in each iteration, assuming the total number of data segments as K and the number of hidden states as N , the total number of multiplications is $(5K - 4)N^2 + (K + 1)N$, the

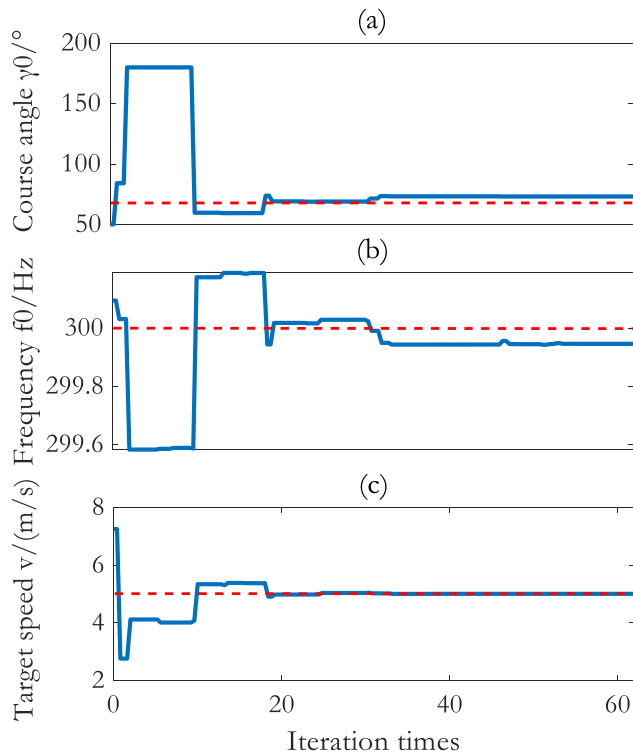


FIG. 10. (Color online) Convergence results for target parameters: (a) course angle, (b) speed, (c) transmit signal frequency.

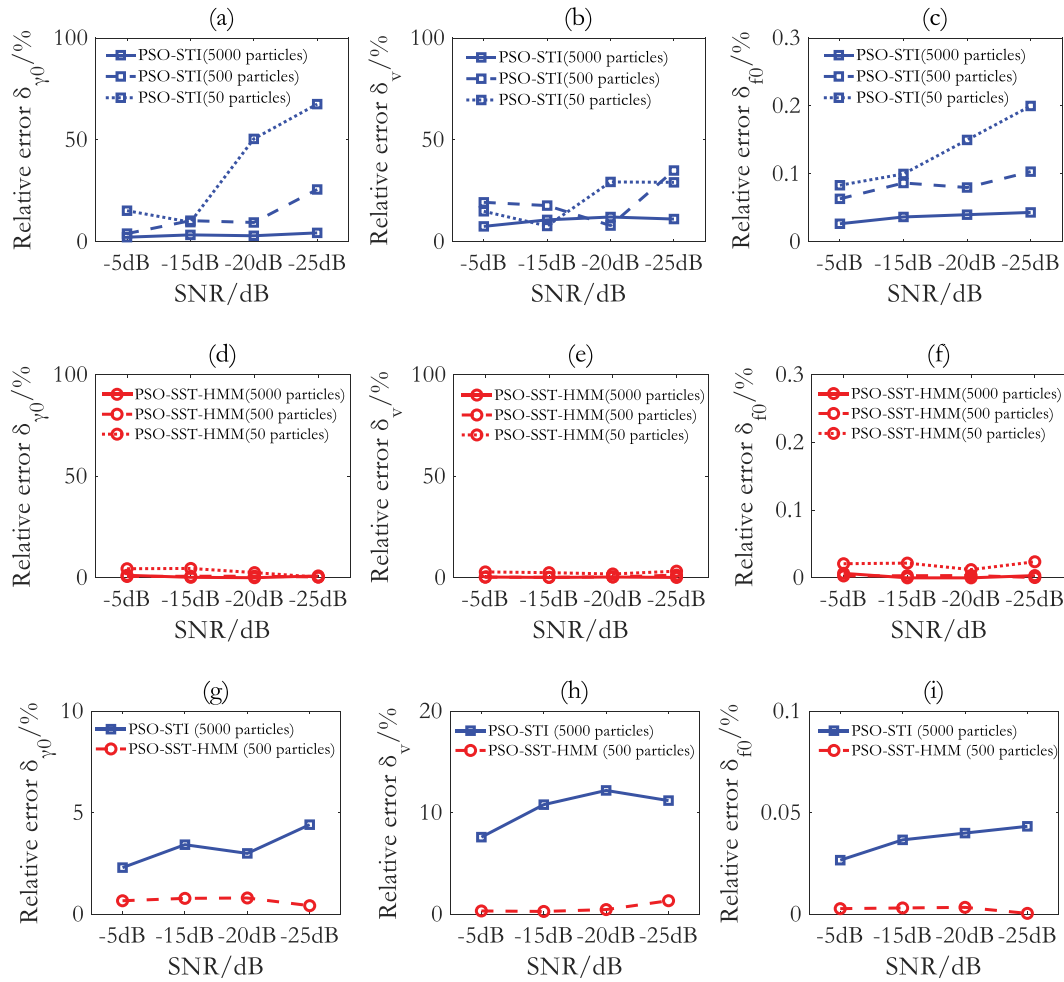


FIG. 11. (Color online) Influence of SNR on the relative error of parameter estimation: (a)–(c) performance of PSO-STI algorithm at different numbers of particles, (d)–(f) performance of PSO-SST-HMM algorithm at different numbers of particles, (g)–(i) performance comparison of two algorithms.

total number of additions is $(3K - 2)N^2 + (2 - 3K)N$, and the total number of assignments is $3KN$.¹⁹ In this study, when implementing the SFJDT algorithm in simulation, no regional segmentation was performed because of the limited frequency range of only 2 Hz and the significant changes in the azimuth of the simulated moving targets. The computational complexity of PSO-SST-HMM and SFJDT in this paper, derived from the above conditions, is shown in Table II. Data were processed on a computer with 1.5 TB random access memory (RAM) and an Intel® Xeon® Platinum

8160 processor (Intel Corporation, Santa Clara, CA) running at 2.10 GHz. For instance, when setting the number of particles to 500, parallel computing is used in each iteration, and up to 48 cores are called. With an SNR of -25 dB, the average number of iterations required for convergence is ca. 60, meaning that $L = 500/48 * 60 = 625$ times trajectory tracking needs to be completed serially. The number of beams, denoted by M , is 361. From Table II, it can be seen that since the value of $(M^2 - L) = 1\,29\,696$ is a large number, under the conditions set herein, the SFJDT algorithm has a larger

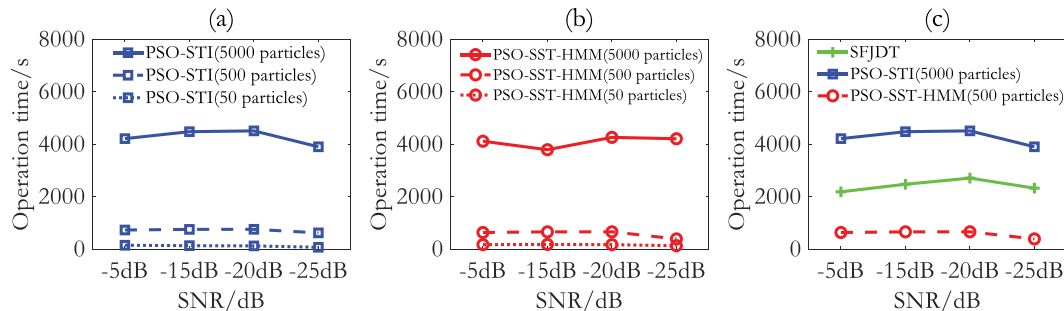


FIG. 12. (Color online) Comparison of operation time for three algorithms: (a) PSO-STI algorithm at different numbers of particles, (b) PSO-SST-HMM algorithm at different numbers of particles, (c) comparison of three algorithms.

TABLE II. Multiplication, addition, and assignment times for algorithms SFJDT and PSO-SST-HMM. Q , the number of frequency points in SFJDT; Q' ($Q' \leq Q$), the number of frequency points in PSO-SST-HMM; M , the number of beams; K , the total number of data segments; L , the number of times the trajectory tracking needs to be completed serially in PSO-SST-HMM.

	Multiplication	Addition	Assignment
PSO-SST-HMM	$(5K - 4)Q'^2L + (K + 1)Q'L$	$(3K - 2)Q'^2L + (2 - 3K)Q'L$	$3KQ'L$
SFJDT	$(5K - 4)Q^2M^2 + (K + 1)QM$	$(3K - 2)Q^2M^2 + (2 - 3K)QM$	$3KQM$
Differences	$(5K - 4)Q^2(M^2 - L) + (K + 1)Q(M - L)$	$(3K - 2)Q^2(M^2 - L) + (2 - 3K)Q(M - L)$	$3KQ(M - L)$

amount of calculation than PSO-SST-HMM. Also, the simulation results show that PSO-SST-HMM is about six times faster than SFJDT.

In conclusion, the proposed method has lower computational complexity due to the introduction of SST, which effectively adds a plane constraint to the signal trajectory tracking problem and reduces the hidden state of HMM from 2D to 1D. In the PSO algorithm, the search for slices in the transformed signal space replaces the search for a 3D trajectory in the original signal space, thereby reducing the number of optimization variables.

D. Key factors affecting algorithm performance

This chapter discusses additional key factors beyond the previously mentioned SNR and the number of particles for PSO in order to provide a more comprehensive evaluation of the algorithm's performance. For the sake of discussion, we consider a target with a course angle of 90° and an emission frequency of 300 Hz. The distance from the array to the CPA point is set to 3 km, with the target's starting and ending points being symmetrical around the CPA point. The step size of the sliding time window for STFT is 30 s. When no changes are specified, there are 27 receivers, a window length of 60 s for the STFT, and a target speed of 5 m/s, covering a trajectory duration of 20 min.

1. Number of receivers and window length for STFT

As is well known, when the spacing between receivers is constant, the number of receivers M in the array and the time window length ΔT used for signal segmentation determine the

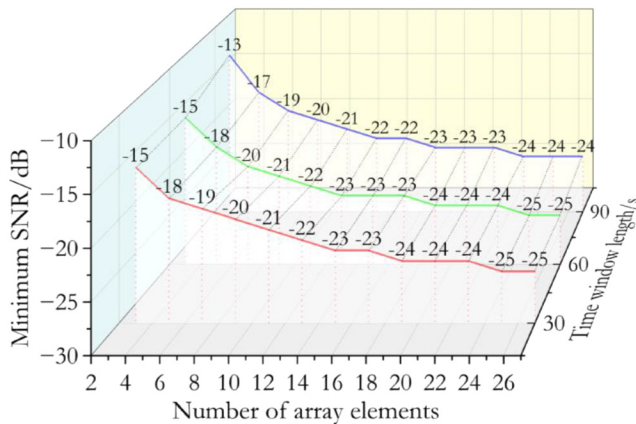


FIG. 13. (Color online) The minimum signal-to-noise ratio for the PSO-SST-HMM algorithm to achieve a 95% trajectory point detection rate with different numbers of receivers and time window lengths.

beam resolution and frequency resolution, respectively. This, in turn, affects the processing gain for beamforming and the FFT, which is given by $10 \log 10(M\Delta T)$. This is the most significant source of gain for the algorithm proposed in this paper. Figure 13 illustrates the minimum SNR required for the proposed algorithm to achieve a trajectory point detection rate of over 95% under different numbers of receivers and window lengths. In this section, the SNR value is rounded to an integer. It can be observed that as the number of receivers increases, the performance of the algorithm steadily improves, allowing for more receivers to be utilized within practical limits. However, the selection of an appropriate time window length is crucial, as it must consider the Doppler effect induced by target motion. Significant frequency variations of the received signal within the time window can lead to spectral spreading, meaning that longer window lengths yield less effective energy concentration in the FFT.

2. Target speed and integration time of the line spectrum trajectory

According to Eq. (4b), when the trajectory of the target motion is the same, the target speed and signal emission frequency determine the rate of frequency change. Coupled with the previously analyzed effects of frequency changes on FFT performance, the proposed algorithm requires a higher SNR as the target speed increases, given the same window length and emission frequency. Additionally, integrating the energy on the LS serves as the second source of gain for the proposed algorithm. Figure 14 presents the minimum SNR required to achieve over 95% trajectory point

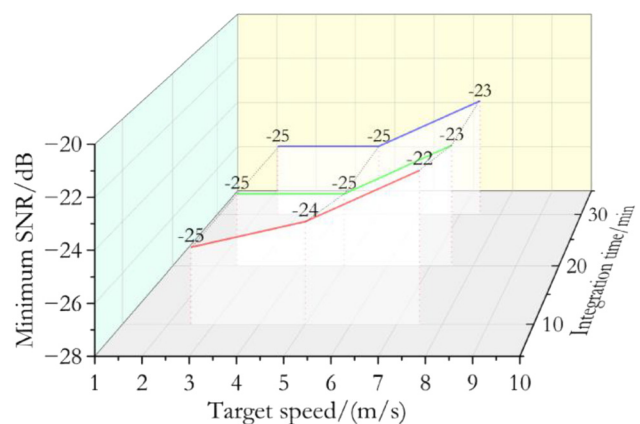


FIG. 14. (Color online) The minimum signal-to-noise ratio for the PSO-SST-HMM algorithm to achieve a 95% trajectory point detection rate at different target speeds and integration times.

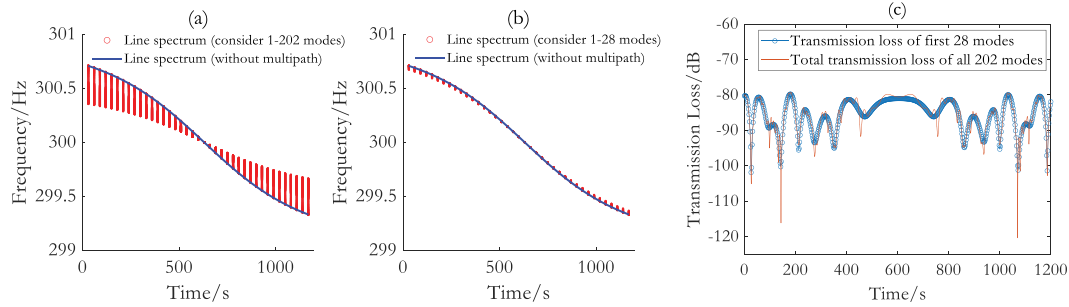


FIG. 15. (Color online) Analysis of transmission loss and Doppler spread considering multipath effects: (a)–(b) Doppler spread of 1–28 and 1–202 normal modes, (c) comparison of transmission loss over time for 1–28 and 1–202 normal modes.

detection rates under different speeds and integration times. It is evident that increased target speed significantly deteriorates the performance of the algorithm, while the gain from integration time is constrained by the beamforming and FFT processing gain required at each point of the FRAZ spectrum when employing the Viterbi algorithm. When the integration time exceeds approximately 20 min, the rate of gain improvement will slow down. Moreover, when the target passes through the same trajectory at different speeds, for example, at a speed of 5 m/s for 10 min and at a speed of 2.5 m/s for 20 min, the algorithm performs better when facing targets with lower speeds.

3. Effects of underwater sound propagation

In real ocean environments, passive sonar receives continuous low-frequency signals from targets, with the long-term data being the result of multipath signal superposition. It is difficult to distinguish signals from different paths in the time domain. To better reflect realistic scenarios, normal mode theory can be employed for simulation and analysis.

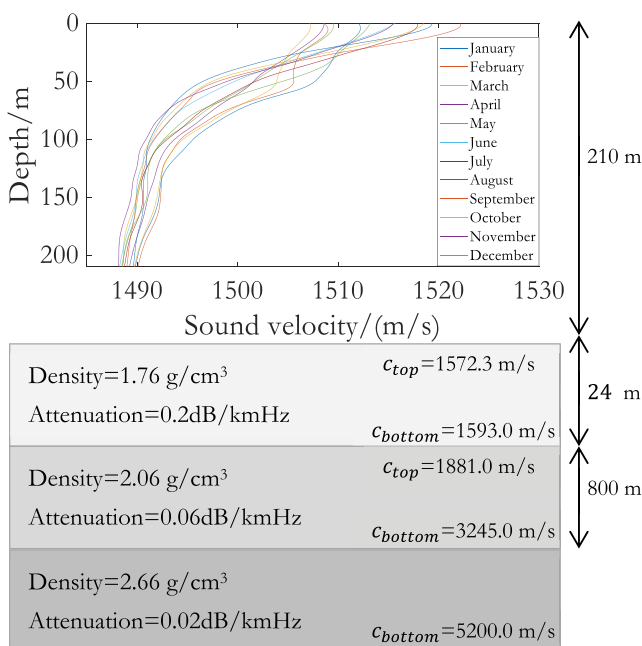


FIG. 16. (Color online) Simulated ocean environment model.

First, the azimuth angles for each order of normal modes are basically the same. Second, the sound waves reflected from the sea surface and seabed usually result in different pitching angles for each order of normal mode, leading to different Doppler frequencies for these modes, as shown in Fig. 15(a). However, as shown in Fig. 15(b), the Doppler frequencies of low-order normal modes, for example, from the 1st to 28th normal modes, have a small difference compared to the case without considering multipath effects. Furthermore, as shown in Fig. 15(c), low-order normal modes are the main component of the received signal. Thus, the spectral broadening caused by multipath effects can be negligible. In summary, the impact of multipath on the measurements of the signal's azimuth and frequency is minimal and can be ignored; the main impact is the amplitude fluctuations of the received signal. Factors, such as the change of sound speed profiles, seabed topography, and target distances can also induce variations in signal amplitude. However, the 60 s FFT and Viterbi algorithm in the proposed method can mitigate amplitude fluctuations to a certain extent.

As shown in Fig. 16, a simulated marine environment was established based on the settings from the Swellex-96 experiment. The ocean noise level was set to 90 dB, and the sound speed profile of the Swellex-96 experimental area from January to December 2019 was calculated using the EN.4 dataset provided by the Met Office Hadley Centre. Figure 17 shows the trajectory point detection rates at different months when the sound source level is 122 dB. It can be seen that the algorithm remains applicable even under situations of fluctuating signal amplitudes, and the impact of

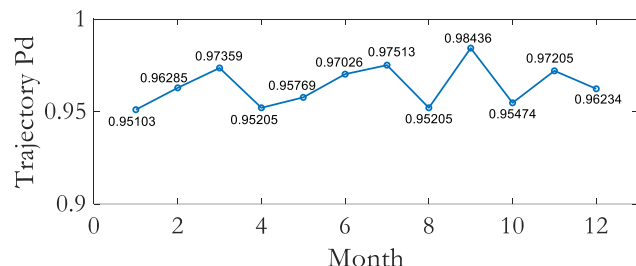


FIG. 17. (Color online) Trajectory point detection rate under varying sound speed profiles from January to December 2019, with the source level at 122 dB and noise level at 90 dB.

variations in sound speed profile in this region on algorithm performance is tolerable.

V. ANALYSIS OF SEA-TRIAL DATA

The algorithm is validated using the SWellEx-96 data obtained in 1996 by the Physical Ocean Laboratory of the University of California, San Diego, CA.²⁵ As shown in Fig. 18, the target was dragged from the start point to the end point at a constant speed of 2.5 m/s. Figure 19 shows the deployment of the seabed horizontal linear array (HLA-North), with its axial direction at 34.5° north. The Global Positioning System data processing reveals that the target course γ_0 relative to the linear array was 245.04° . Experimental data processing was focused on tonal signals with four different sound source levels emitted by the experimental vessel within 32 min. The source levels (SL) of emission sound are 158, 132, 128, and 124 dB, and the emission frequencies are 64, 67, 70, and 73 Hz, respectively. At present, there has been almost no separate use of single-frequency signals at sound levels of 128 dB and 124 dB for joint long-term tracking of frequency and azimuth in the processing of SWellEx-96 data.^{26,27} This limitation arises from the very low SNR.

Figure 20 shows the tracking results obtained using three different algorithms for four LS trajectories (64, 67, 70, 73 Hz). For LS tracking at the center frequencies of 64, 67, and 70 Hz, data accumulation was performed within 0–32 min.

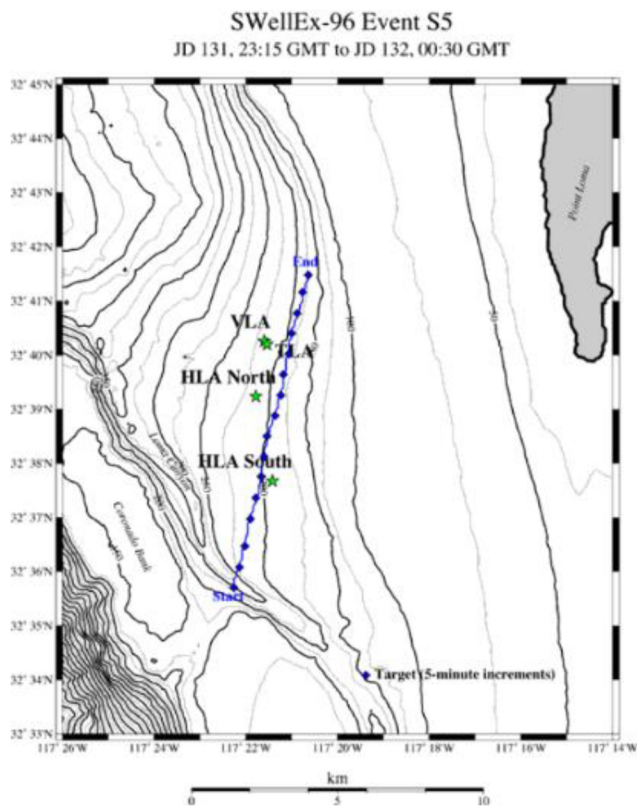


FIG. 18. (Color online) S5 experimental latitude and longitude map. Reprinted with permission from Murray and Ensberg, swellex96.ucsd.edu (Ref. 25).

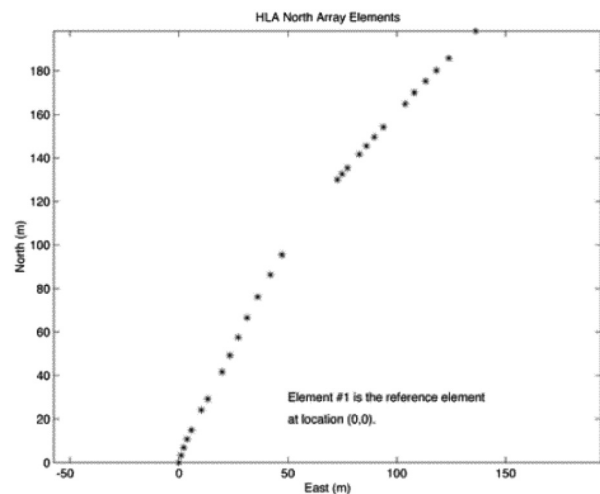


FIG. 19. Deployment of HLA-North. Reprinted with permission from Murray and Ensberg, swellex96.ucsd.edu (Ref. 25).

However, for the 73 Hz LS, the low SNR during 0–14 min and 26–32 min affected the trajectory tracking and accumulation processes. Consequently, only the data for 15–25 min were considered in this case. The trajectory point detection probabilities of the three algorithms are presented in Table III.

The SFJDT algorithm tracks within the 2 Hz bandwidth centered on each transmission frequency, and the resulting trajectory points are represented by the symbol “+” in Fig. 20. This method effectively tracks the trajectory at 64 and 67 Hz, with only a few points affected by noise and deviating from the real trajectory. However, it fails to track the LS of 70 and 73 Hz with lower SNRs.

According to the simulation results, the PSO-STI algorithm used 5000 particles in the parameter optimization process and achieved a 100% trajectory detection rate at high source levels (64, 67 Hz), which is the highest among the three algorithms. The reason is that during the movement of the target, the amplitude of the received signal fluctuates due to factors such as multipath interference, changes in distance between the target and receiver, or the suspension of signal emission by the target during certain periods. The STI algorithm always integrates along a fixed coordinate trajectory corresponding to a hypothesized target track and does not “lock on” to a strong noise peak or “lose lock” during a signal fade, as a recursive signal-follower might.¹⁵ As depicted in Fig. 21(a), the signal experiences a sudden fade at ca. 700 s (perhaps the target has paused transmitting signals at this time), impacting the trajectory tracking of PSO-SST-HMM and SFJDT, while PSO-STI remains unaffected. However, at low SNR, the high parameter dimension makes it difficult for the PSO-STI to find the optimal value, failing to track the trajectory accurately.

The PSO-SST-HMM algorithm proposed herein uses 500 particles, although the trajectory point detection rate is not very high (69.84, 69.84, 58.73, 63.16%) when tracking the four line spectra, which is due to the impact of received signal amplitude fluctuations. However, the method proposed in this paper mainly aimed at LS tracking under extremely low SNR.

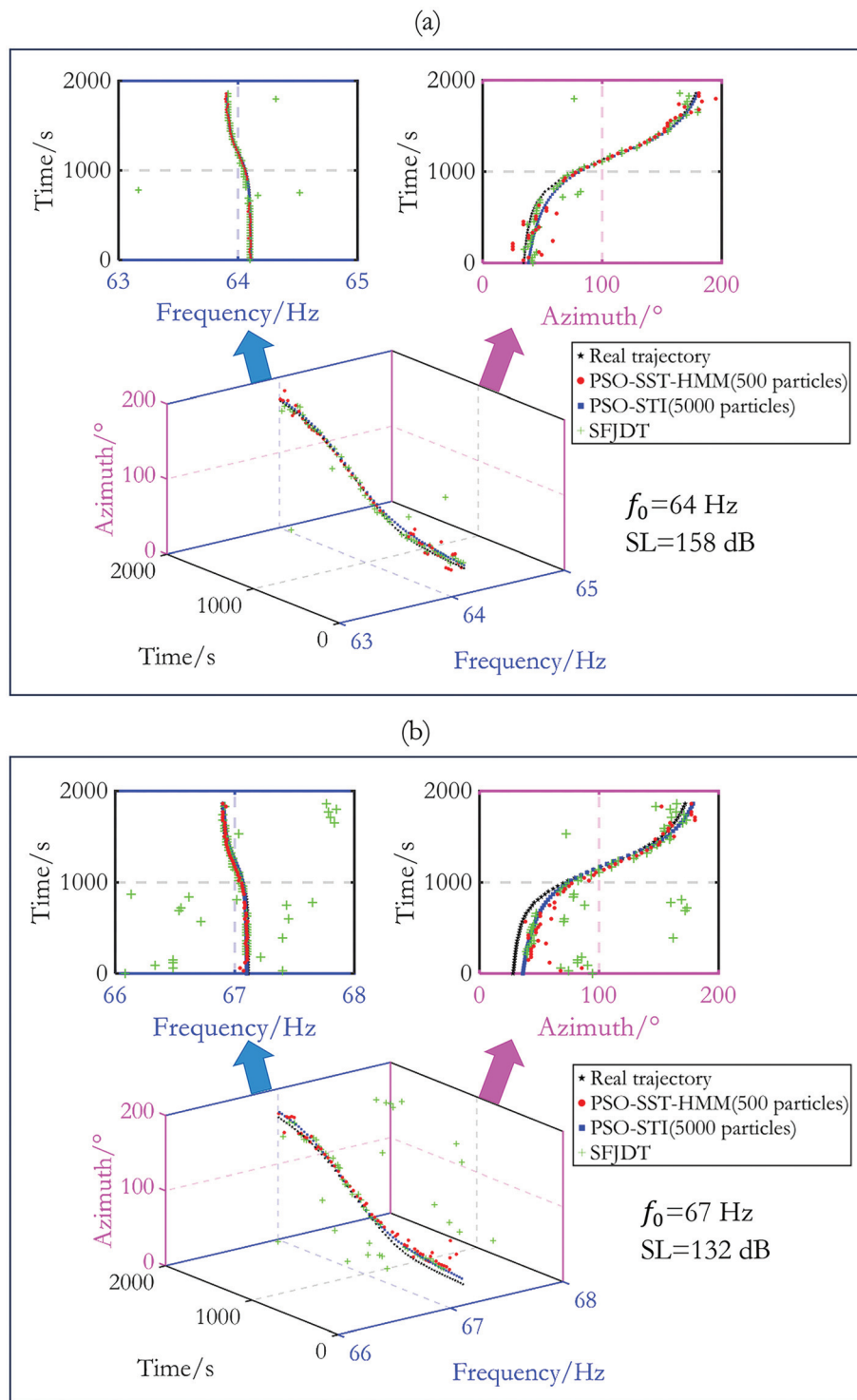


FIG. 20. Tracking results of four LS trajectories with different source levels using three different algorithms: (a) $f = 64$ Hz; $SL = 158$ dB, (b) $f = 67$ Hz; $SL = 132$ dB, (c) $f = 70$ Hz; $SL = 128$ dB, (d) $f = 73$ Hz; $SL = 124$ dB.

It is a kind of TBD algorithm that focuses on tracking. Unlike the DBT algorithm, it focuses on global tracking of the LS within the processing time range, and individual points that are not tracked do not affect overall performance. From the experimental results, incomplete tracking of trajectory points does not affect finding the correct slice, nor does it affect the

estimation of the three parameters of the target, and the trajectory formed by the tracked points is also basically consistent with the real trajectory.

The tracking effect is better in frequency, and the azimuth angle may be easily affected by the wider beam width in the end fire direction of beamforming, resulting in a

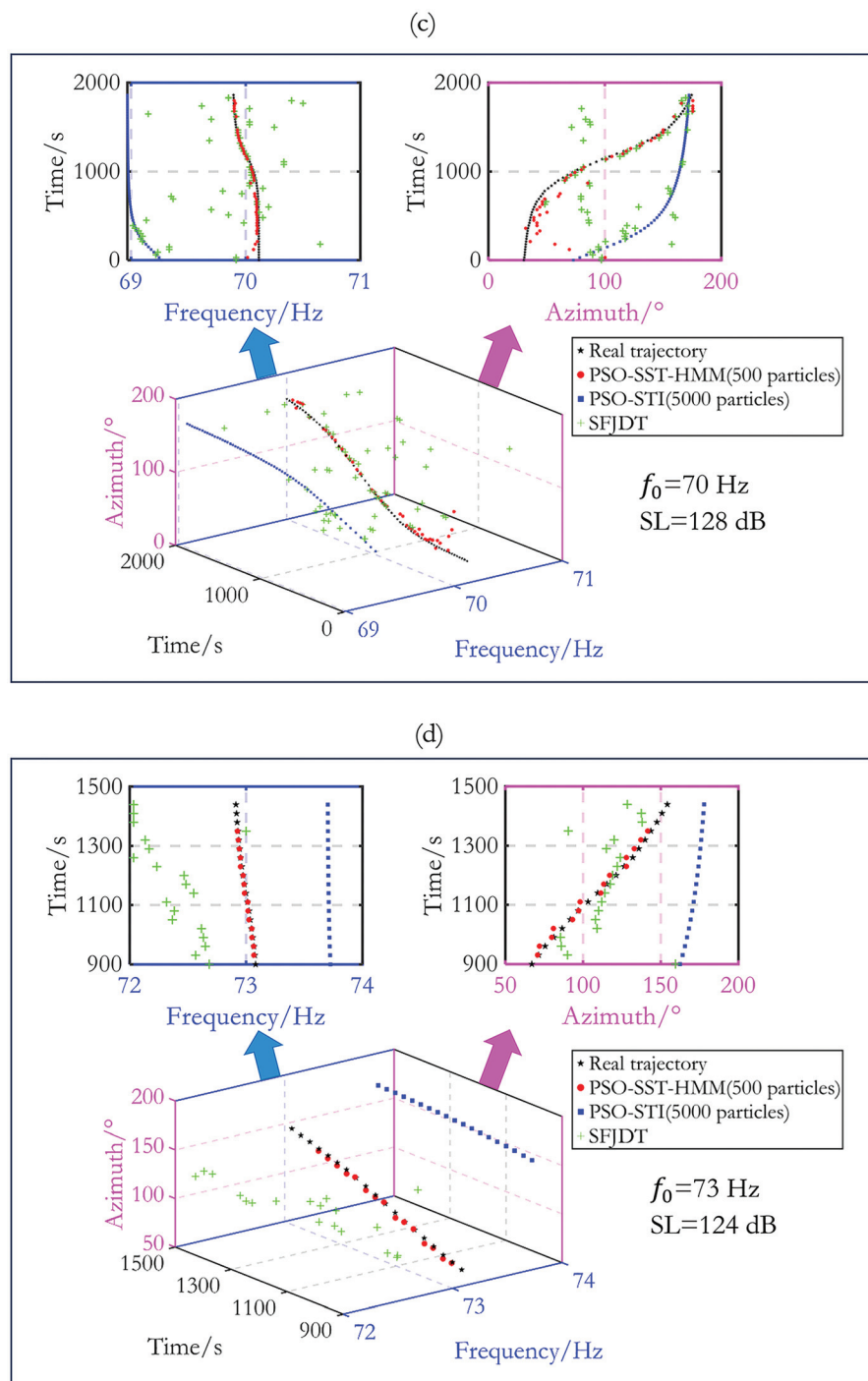


FIG. 20. (Color online) (Continued)

TABLE III. Trajectory point detection rates of three algorithms under different source levels.

	64 Hz (158 dB)	67 Hz (132 dB)	70 Hz (128 dB)	73 Hz (124 dB)
PSO-SST-HMM	69.84	69.84	58.73	63.16
PSO-STI	100.00	100.00	0.00	0.00
SFJDT	63.49	61.90	33.33	0.00

TABLE IV. Parameter optimization results of PSO-SST-HMM algorithm at different source levels. $\hat{\gamma}_0$, estimated target course; δ_{γ_0} , relative error of target course; $\hat{v}$, estimated target speed; δ_v , relative error of target speed; $\hat{f}_0$, estimated frequency of transmitted signal; δ_{f_0} , relative error of transmitted signal frequency.

	$\hat{\gamma}_0$ (°)	δ_{γ_0} (%)	$\hat{v}$ (m/s)	δ_v (%)	$\hat{f}_0$ (Hz)	δ_{f_0} (%)
64 Hz (158 dB)	246.50	0.59	2.49	0.29	64.00	0.00
67 Hz (132 dB)	252.76	3.15	2.44	2.35	67.00	0.01
70 Hz (128 dB)	250.26	2.13	2.29	8.31	70.00	0.00
73 Hz (124 dB)	251.14	2.49	2.48	0.76	72.99	0.01

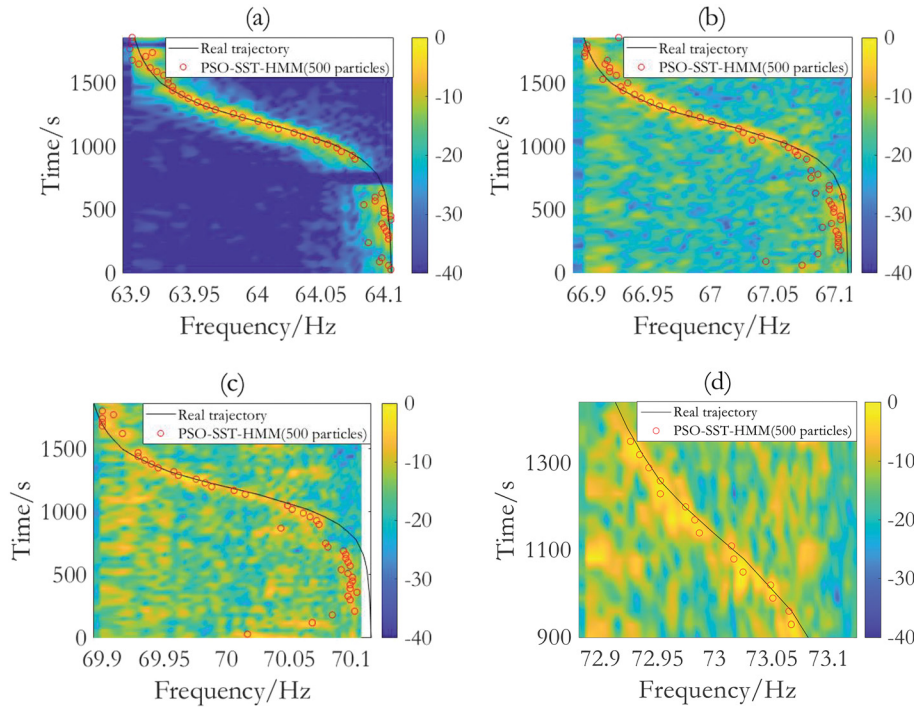


FIG. 21. (Color online) Tracking trajectory on searched slices using PSO-SST-HMM algorithm under different source levels: (a) $f = 64$ Hz; $SL = 158$ dB, (b) $f = 67$ Hz; $SL = 132$ dB, (c) $f = 70$ Hz; $SL = 128$ dB, (d) $f = 73$ Hz; $SL = 124$ dB.

relatively large tracking offset. In addition, the trajectory detection rate of the 73 Hz LS surpasses that of the 70 Hz LS, primarily because we specifically chose a time segment with the highest SNR of the 73 Hz LS for calculations. For the LS of 73 Hz, the SNR is very low during time intervals except for 15–25 min, mainly influenced by the distance between the target and the array. Without segmenting the signal and focusing on the period with the highest SNR, effective tracking of the LS will not be achieved. In practical applications, it is recommended to try different time periods according to the specific situation. Table IV presents the relative error in parameter estimation for the PSO-SST-HMM algorithm, rounded to two decimal places. Under all four SNR conditions, the estimation results for all three parameters consistently exhibit high accuracy. Notably, the relative error in estimating the emission frequency is the lowest, corresponding to the highest accuracy requirement for parameter f_0 in slice search, as mentioned in Ref. 21. Figure 21 shows the trajectory tracking of the proposed algorithm on the slice. The solid line represents the real trajectory, and the circles denote the trajectory points tracked by the PSO-SST-HMM algorithm. Experimental results demonstrate the superior tracking performance of the proposed PSO-SST-HMM algorithm, particularly in low-SNR conditions.

VI. CONCLUSIONS

In this paper, an LS trajectory tracking method based on SST is proposed, which is designed to track the weak low-frequency LS components in signals emitted by a target moving at a constant speed along a straight line. PSO is used to search for three parameters of the target motion,

including the frequency of the transmitted signal, target speed, and course angle. These three parameters are used to transform the FRAZ-T signal space and extract a slice containing the target trajectory. Then, the Viterbi algorithm is used to track trajectory points on that slice, and the energy of effective points is accumulated as the cost function for parameter optimization.

The proposed algorithm constrains the trajectory tracking to a plane through SST, which utilizes the linear relationship between the frequency of the received signal and the cosine value of relative bearing when the target moves at a constant speed. Then, the number of parameters to be optimized in PSO and the hidden state of HMM is reduced, thereby reducing system complexity, mitigating the impact of the noise, and improving the ability of LS tracking.

The simulation results demonstrate that the proposed method outperforms PSO-STI and SFJDT algorithms regarding trajectory tracking, parameter estimation, and computing efficiency under the conditions specified in this study. Specifically, under low SNR conditions, when an appropriate number of particles is selected, the relative error between the estimated and actual parameters remains at a low level, and the trajectory point detection rate is higher than other tracking algorithms. Moreover, when using the PSO algorithm, the proposed method requires fewer particles than the PSO-STI algorithm, leading to computational time reduction.

By applying the proposed algorithm to process the SWellEx-96 S5 experimental data and carefully selecting the appropriate bandwidth and time interval, it can be seen that tracking an LS with a sound source level of 124 dB is feasible. On the contrary, the PSO-STI and SFJDT algorithms can only track LS signals with sound source levels of 158 and 132 dB when processing data in the 2 Hz bandwidth.

It seems that the PSO-SST-HMM algorithm offers distinct advantages without relying on excessive ocean environmental parameter information.

The proposed method still has certain limitations. First, it can only deal with situations where there is only one tonal signal within the processing bandwidth. However, there are many line spectrums (LSs) in actual ocean signals, so it is necessary to use this algorithm in multiple narrow bands simultaneously. Usually, only the low-frequency component of the signal radiated by the marine target can achieve long-distance propagation, and different types of targets have specific bandwidths, so the required search is limited and feasible. Second, it is specifically designed for a single target with a constant speed. Future work should attempt to overcome the difficulties brought by multi-target situations and more complex motion patterns.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Ethics Approval

This research adheres to the principles of research integrity and scientific rigor. The data and methods employed in this study are in accordance with ethical standards and guidelines relevant to the field of research.

DATA AVAILABILITY

The experimental data used in this study are available at the following website: <https://swellex96.ucsd.edu/>. Permission to use these data has been obtained.

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